National University of Computer and Emerging Sciences

FAST School of Computing **Fall-2022** **Islamabad Campus**

MT-1004: Linear Algebra
BS-CY (A, B, C & D)

Wednesday, 9th November, 2022

Course Instructor

Sehrish Hassan Shigri

Serial No:

Sessional Exam-II
Total Time: 1 Hour
Total Marks: 60

_____
Signature of Invigilator

_____ _____ _____ _____
Student Name Roll No. Section Signature

DO NOT OPEN THE QUESTION BOOK OR START UNTIL INSTRUCTED.

Instructions:

1. Attempt on question paper. Attempt all of them. Read the question carefully, understand the question, and then attempt it.
2. No additional sheet will be provided for rough work. Use the back of the last page for rough work.
3. If you need more space write on the back side of the paper and clearly mark question and part number etc.
4. After asked to commence the exam, please verify that you have **ten (10)** different printed pages including this title page. There are a total of **3** questions.
5. Calculator sharing is strictly prohibited.
6. Use permanent ink pens only. Any part done using soft pencil will not be marked and cannot be claimed for rechecking.

	Q-1	Q-2	Q-3	Total
Marks Obtained				
Total Marks	**17**	**20**	**23**	**60**

Question 1 [10+7 Marks]

Part A (10 Marks)

Let $T: \mathbb{R}^2 \to \mathbb{R}^2$ be a composition of linear transformations that first rotates points through $-2\pi/3$ radians, then reflects points through the line $x_2 = -x_1$ and then projects points through x_1-axis. Find the standard matrix A of this transformation T and find the image of $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$ under T. (Show the transformation matrix of each transformation as well).

T_1: Rotation through $-2\pi/3$

$$A_1 = \begin{bmatrix} T_1(\vec{e}_1) & T_1(\vec{e}_2) \end{bmatrix}$$

$$T_1(\vec{e}_1) = \begin{bmatrix} \cos -2\pi/3 & -\sin -2\pi/3 \\ \sin -2\pi/3 & \cos -2\pi/3 \end{bmatrix} \vec{e}_1$$

$$= \begin{bmatrix} -1/2 & \sqrt{3}/2 \\ -\sqrt{3}/2 & -1/2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} -1/2 \\ -\sqrt{3}/2 \end{bmatrix}$$

$$T_1(\vec{e}_2) = \begin{bmatrix} \sqrt{3}/2 \\ -1/2 \end{bmatrix}$$

$$A_1 = \begin{bmatrix} -1/2 & \sqrt{3}/2 \\ -\sqrt{3}/2 & -1/2 \end{bmatrix}$$

T_2: Reflection through the line $x_2 = -x_1$

$$T_2\left(\begin{bmatrix} -1/2 \\ -\sqrt{3}/2 \end{bmatrix}\right) = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} -1/2 \\ -\sqrt{3}/2 \end{bmatrix}$$

$$= \begin{bmatrix} \sqrt{3}/2 \\ 1/2 \end{bmatrix}$$

$$T_2\left(\begin{bmatrix} \sqrt{3}/2 \\ -1/2 \end{bmatrix}\right) = \begin{bmatrix} 1/2 \\ -\sqrt{3}/2 \end{bmatrix}$$

$$A_2 = \begin{bmatrix} \sqrt{3}/2 & 1/2 \\ 1/2 & -\sqrt{3}/2 \end{bmatrix}$$

T_3: Projection onto x_1-axis

$$T_3\begin{bmatrix} \sqrt{3}/2 \\ 1/2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \sqrt{3}/2 \\ 1/2 \end{bmatrix}$$

$$= \begin{bmatrix} \sqrt{3}/2 \\ 0 \end{bmatrix}$$

$$T_3\begin{bmatrix} 1/2 \\ -\sqrt{3}/2 \end{bmatrix} = \begin{bmatrix} 1/2 \\ 0 \end{bmatrix}$$

Hence,

$$A = \begin{bmatrix} \sqrt{3}/2 & 1/2 \\ 0 & 0 \end{bmatrix}$$

Image of $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$ under T is

$$T\left(\begin{bmatrix} 1 \\ -1 \end{bmatrix}\right) = \begin{bmatrix} \sqrt{3}/2 & 1/2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} (\sqrt{3}+1)/2 \\ 0 \end{bmatrix}$$

Part B (5+2 Marks)

Let $A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 2 & 4 \\ 1 & 2 & 3 \end{bmatrix}$ be a transformation matrix for a linear transformation T.

(i) Find inverse of A using row operations, if possible.

(ii) Is T one-to-one, onto, or both? Explain your Answer.

a. $\begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 4 \\ 0 & 0 & 1 \end{bmatrix}$ $R_3 \leftrightarrow R_1$

$= \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$ $R_2 / 2$

$= \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$ $R_1 - 2R_2$

$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$ $R_1 + R_3$

$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ $R_2 - 2R_3$

$E_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ $R_3 \leftrightarrow R_1 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$

$E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ $R_1 - 2R_2 = \begin{bmatrix} 1 & -2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$E_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ $R_1 + R_3 = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$I = E_3 E_2 E_1 A$

$E_1^{-1} E_2^{-1} E_3^{-1} = A$

$E_1^{-1} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$, $E_2^{-1} = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$E_3^{-1} = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

b. It is both because the ~~the~~ matrix has pivot in every column as shown in part a.

Question 2 [10+5+5 Marks]

Part A (10 Marks)

In $\mathbb{P}_1$, find the change-of-coordinates matrix from the basis $T = [t-1,\ t+1]$ to the basis $S = [t,\ t-3]$. Then find the S-coordinate vector for $5t+1$.

To compute $P_{S\leftarrow T}$, we need to solve the vector equations

$$a_1\mathbf{v}_1 + a_2\mathbf{v}_2 = \mathbf{w}_1$$
$$b_1\mathbf{v}_1 + b_2\mathbf{v}_2 = \mathbf{w}_2$$

simultaneously by transforming the resulting partitioned matrix

$$\left[\begin{array}{cc|c|c} 1 & 1 & 1 & 1 \\ 0 & -3 & -1 & 1 \end{array}\right]$$

to reduced row echelon form. The result is (verify)

$$\left[\begin{array}{cc|c|c} 1 & 0 & \frac{2}{3} & \frac{4}{3} \\ 0 & 1 & \frac{1}{3} & -\frac{1}{3} \end{array}\right],$$

so

$$P_{S\leftarrow T} = \begin{bmatrix} \frac{2}{3} & \frac{4}{3} \\ \frac{1}{3} & -\frac{1}{3} \end{bmatrix}.$$

If $\mathbf{v} = 5t+1$, then expressing $\mathbf{v}$ in terms of the T-basis, we have

$$\mathbf{v} = 5t+1 = 2(t-1) + 3(t+1),$$

so $[\mathbf{v}]_T = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$. Then

$$[\mathbf{v}]_S = P_{S\leftarrow T}[\mathbf{v}]_T = \begin{bmatrix} \frac{2}{3} & \frac{4}{3} \\ \frac{1}{3} & -\frac{1}{3} \end{bmatrix}\begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} \frac{16}{3} \\ -\frac{1}{3} \end{bmatrix}.$$

Computing $[\mathbf{v}]_S$ directly, we find that

$$\mathbf{v} = 5t+1 = \tfrac{16}{3}t - \tfrac{1}{3}(t-3), \quad \text{so } [\mathbf{v}]_S = \begin{bmatrix} \frac{16}{3} \\ -\frac{1}{3} \end{bmatrix}.$$

Hence

$$[\mathbf{v}]_S = P_{S\leftarrow T}[\mathbf{v}]_T.$$

Part B (5 Marks)

Consider the 5×4 matrix

$$A = [\mathbf{v}_1 \quad \mathbf{v}_2 \quad \mathbf{v}_3 \quad \mathbf{v}_4].$$

The vector $\begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$ is in Nul A. Write $\mathbf{v}_4$ as a linear combination of $\mathbf{v}_1$, $\mathbf{v}_2$, $\mathbf{v}_3$.

The fact that $\begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$ is in $\ker(A)$ means that

$$A \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} = [\vec{v}_1 \ \vec{v}_2 \ \vec{v}_3 \ \vec{v}_4] \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} = \vec{v}_1 + 2\vec{v}_2 + 3\vec{v}_3 + 4\vec{v}_4 = \vec{0}, \text{ so that } \vec{v}_4 = -\tfrac{1}{4}\vec{v}_1 - \tfrac{1}{2}\vec{v}_2 - \tfrac{3}{4}\vec{v}_3.$$

Part C (5 Marks)

Find a basis for the span(S) as a subspace of $\mathbb{R}^3$ where $S = \left\{ \begin{bmatrix} -2 \\ 1 \\ 3 \end{bmatrix}, \begin{bmatrix} 4 \\ -1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ 5 \end{bmatrix} \right\}$.

Since $\begin{bmatrix} 2 \\ 0 \\ 5 \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \\ 3 \end{bmatrix} + \begin{bmatrix} 4 \\ -1 \\ 2 \end{bmatrix}$, the vectors in S are linearly dependent. Since the first two vectors are not scalar multiples of each other a basis for $\mathbf{span}(S)$ is $\left\{ \begin{bmatrix} -2 \\ 1 \\ 3 \end{bmatrix}, \begin{bmatrix} 4 \\ -1 \\ 2 \end{bmatrix} \right\}$.

Question 3 [6+5+12 Marks]
Part A (2+2+2 Marks)

i) Let U be the set of all linear transformations of V into W and let $O: V \rightarrow W$ be the zero linear transformation defined by $O(\boldsymbol{x}) = \mathbf{0}_W$ for all $\boldsymbol{x}$ in V. Describe the Ker O.

The $Ker\ O = \{\boldsymbol{x} \in V | O(\boldsymbol{x}) = \mathbf{0}_W\} = V$

ii) Consider an $m \times n$ matrix A and what are often called the fundamental subspaces determined by A. Which of the subspaces Row A, Col A, Nul A, Row A^T , Col A^T, and Nul A^T are in $\mathbb{R}^m$ and which are in $\mathbb{R}^n$? How many distinct subspaces are in this list?

The matrix A maps $\mathbb{R}^n$ to $\mathbb{R}^m$, therefore,
Row $A \in \mathbb{R}^n$, Col $A \in \mathbb{R}^m$, $Nul\ A \in \mathbb{R}^n$, Row $A^T \in \mathbb{R}^m$, Col $A^T \in \mathbb{R}^n$, and Nul $A^T \in \mathbb{R}^m$
There are four distinct subspaces,
Row A= Col A^T, $Nul\ A \in \mathbb{R}^n$
Col A = Row A^T, Nul $A^T \in \mathbb{R}^m$

iii) Let $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation.

1. What is the dimension of the range of T if T is a one-to-one mapping? Explain.
   Since there is no free variable because T is one-to-one, the
   $$\dim(range\ T) = n$$

2. What is the dimension of the kernel of T if T maps $\mathbb{R}^n$ onto $\mathbb{R}^m$? Explain.

Since the transformation is onto, this means there are m pivot positions, this means $m \leq n$
So,
$$\dim(\ker T) = n - m$$

Part B (5 Marks)

Let S and T be the subspaces of $\mathbb{R}^4$ defined by

$$S = \left\{ \begin{bmatrix} s \\ t \\ 0 \\ 0 \end{bmatrix}, s, t \in \mathbb{R} \right\} \text{ and } T = \left\{ \begin{bmatrix} 0 \\ s \\ t \\ 0 \end{bmatrix}, s, t \in \mathbb{R} \right\}$$

Find dim(S), dim(T), and dim(S ∩ T).

Since $S = \left\{ s \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} \middle| \; s, t \in \mathbb{R} \right\}$, then $\dim(S) = 2$. Since $T = \left\{ s \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \middle| \; s, t \in \mathbb{R} \right\}$, we also have that $\dim(T) = 2$. For the intersection, since

$$S \cap T = \left\{ s \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} \middle| \; s \in \mathbb{R} \right\}, \text{ then } \dim(S \cap T) = 1.$$

Part C (5 +2+2+1+1+1 Marks)

Let $L: \mathbb{P}_2 \to \mathbb{P}_3$ be the linear transformation defined by $L(p(t)) = t^2 p'(t)$.

i) Show that L is a linear transformation.

Let $p(t), q(t) \in \mathbb{P}_2$ and $x, y \in \mathbb{R}$ such that $p(t) = a_0 + a_1 t + a_2 t^2$ and $q(t) = b_0 + b_1 t + b_2 t^2$,
We need to show that

$$L\big(xp(t) + yq(t)\big) = xL\big(p(t)\big) + yL\big(q(t)\big).$$

Consider,

$$L\big(xp(t) + yq(t)\big) = L(xa_0 + xa_1 t + xa_2 t^2 + yb_0 + yb_1 t + yb_2 t^2)$$
$$= t^2\big((xa_1 + yb_1) + 2(xa_2 + yb_2)t\big) = (xa_1 + yb_1)t^2 + 2(xa_2 + yb_2)t^3 \qquad (1)$$

Now,

$$xL\big(p(t)\big) + yL\big(q(t)\big) = xt^2(a_1 + 2a_2 t) + yt^2(b_1 + 2b_2 t)$$
$$= (xa_1 + yb_1)t^2 + 2(xa_2 + yb_2)t^3 \qquad (2)$$

From (1) and (2), we get

$$L\big(xp(t) + yq(t)\big) = xL\big(p(t)\big) + yL\big(q(t)\big).$$

ii) Find a basis for Ker$L$.

$$\ker L = \{p(t) = a_0 + a_1 t + a_2 t^2 | L(p(t)) = 0\}$$
$$= \{p(t) = a_0 + a_1 t + a_2 t^2 | t^2(a_1 + 2a_2 t) = 0\}$$
$$= \{p(t) = a_0 + a_1 t + a_2 t^2 | a_1 = 0, a_2 = 0\}$$
$$= \{a_0 | a_0 \in \mathbb{R}\} = span\,\{1\}$$

Therefore, the basis is $\{1\}$

iii) Find the basis for range L.

$$range\ L = \{L(p(t)) | p(t) = a_0 + a_1 t + a_2 t^2\}$$
$$= \{t^2(a_1 + a_2 t) | p(t) = a_0 + a_1 t + a_2 t^2\}$$
$$= \{a_1 t^2 + a_2 t^3 | a_1, a_2 \in \mathbb{R}\}$$

iv) Is this transformation one-to-one? Explain your answer.

No, since there are infinitely many vectors in $\mathbb{P}_2$ that map to $0 \in \mathbb{P}_3$.

v) Is this transformation onto? Explain your answer.

No, since no polynomial of the form $a_0 \in \mathbb{P}_3$ does not have any pre-image in $\mathbb{P}_2$ under L.

vi) Is this an isomorphism? Why?

No, because an isomorphism is a linear transformation that is both one-to-one and onto.

Rough Sheet